Ofqual QAN	**600/0406/X**
Quartz ID	**1156 / 1157**
Qualification	**Level 4 Diploma in Computing / Level 4 Diploma in Computing with Business Management**

Assessment	**Databases 2.0**
Assessment Type	**Examination**
Date	**Thursday 27th February 2025**
Time	**09.00am – 12.00pm**

Mark Scheme

This marking scheme has been prepared as a **guide only** to markers. This is not a set of model answers, or the exclusive answers to the questions, and there will frequently be alternative responses which will provide a valid answer. Markers are advised that, unless a question specifies that an answer be provided in a particular form, then an answer that is correct (factually or in practical terms) **must** be given the available marks.

If there is doubt as to the correctness of an answer, the relevant NCC Education materials should be the first authority.

Throughout the marking, please credit any valid alternative point.

Answer ALL questions

Marks

Question 1

In a hospital system, data is maintained about patients, their treatments, the doctors attending to them and the wards where they are placed. A patient can undergo various treatments and each treatment might involve one or more doctors. Doctors are assigned to specific wards, with each ward hosting multiple treatments.

a) Identify **FOUR (4)** entities in the hospital database system based on the scenario provided. **4**

MARK SCHEME:

Includes:

- ***patient***
- ***treatment***
- ***doctor***
- ***ward***

(1 mark per bullet point)
Any other suitable answers should be accepted.

b) Explain what is meant by propagation constraints in a database system and identify **TWO (2)** key features of propagation constraints. **4**

MARK SCHEME:

Propagation constraints determine how changes in one table (1 mark) affect to related tables (1 mark). They ensure referential integrity is maintained when records are updated or deleted (1 mark).

Award 1 mark for any of the points above, up to a max of 2 marks.

Features:

Cascade (1 mark) and Restrict (1 mark).

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c) Provide **ONE (1)** example of how propagation constraints could be applied in the hospital database. **2**

MARK SCHEME:

If a patient is discharged and removed from the database, propagation constraints can ensure that all associated treatments are also removed (cascade) or prompt a warning if related data exists (restrict).

1 mark for the situation, 1 mark for applying either cascade or restrict.

Total 10 Marks

Question 2

Revisit the hospital database.

Data integrity must be maintained when recording critical operations like admitting a patient, assigning a treatment or updating the availability of doctors.

For example, when a patient undergoes surgery, multiple updates need to happen simultaneously, such as updating patient records, logging treatment details and updating the status of the operation room.

a) Identify each property in **ACID**. Provide **ONE (1)** practical example of how each property would help ensure smooth operations for the hospital database system. **8**

MARK SCHEME:

Atomicity (2 marks):
Ensures that all steps in a transaction are completed successfully or none at all. For instance, when admitting a patient, the database must either update all related records (patient details, doctor assignment, ward allocation) or roll back completely if any part fails.

Consistency (2 marks):
Guarantees that a transaction brings the database from one valid state to another. For example, after a patient is discharged, the database ensures that the patient is no longer linked to any treatments or assigned to a ward.

Isolation (2 marks):
Prevents concurrent transactions from interfering with each other. For example, if one doctor is assigning treatments while another updates patient records, isolation ensures their actions do not overlap or corrupt data.

Durability (2 marks):
Ensures that once a transaction is completed, its effects are permanently saved, even in case of a system crash. For example, once a patient's treatment record is updated, that change remains in the database regardless of power failures.

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1 mark for each property, 1 mark for an example, up to a max of 8 marks.

b) Explain the purpose of the **ROLLBACK** command in SQL. Identify a situation where it would be used. **2**

MARK SCHEME:

When a patient undergoes surgery:

The ROLLBACK command undoes all changes in the current transaction (1 mark). Use case: If a nurse accidentally deletes a patient record while entering new data, a rollback can restore the database to its previous state (1 mark).

Total 10 Marks

Question 3

10

Scenario:

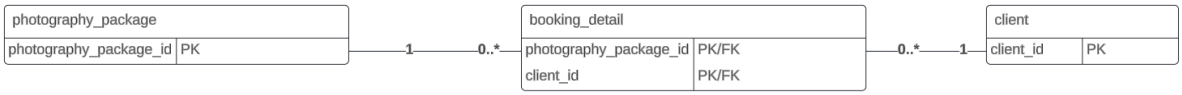
A photography studio wants to create a database to manage its clients, photography packages and bookings.

Each client can book one or more photography packages, and each photography package can be booked by multiple clients.

Draw an Entity Relationship Diagram for the photography database.

You need to identify relationships, cardinalities and label all the primary **and** foreign keys. Clearly state which entity the keys belong to.

MARK SCHEME:



1 mark for all relationships and 1 mark each for cardinalities. The notation types should not be the focus of this marking, up to a max of 4 marks.

1 mark for each correctly labelled PK and FK, up to a maximum of 6 marks. A booking_detail_id as PK should be accepted but must be along with booking_id and photography_package_id as FK.

Total 10 Marks

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Marks

Question 4

a) Flex Gym requires staff to complete the following form when members sign up to their workout plans. **8**

You are required to begin the normalisation process **and** list **EIGHT (8)** attributes that would be suitable for managing Flex Gym memberships in a database system.

Flex Gym Membership Form

Date: 15 January 2025

Member ID: M301

Member Name: Tom Johnson

Workout Plans Enrolled:

- **Strength Training** (Trainer: Luke, Duration: 6 Weeks)
- **HIIT** (Trainer: Daniel, Duration: 4 Weeks)

Personal Trainer Name: Luke

Membership Type: Gold

Membership Start Date: 01 July 2024

Payment Type: Cash/Debit or Credit Card: **CREDIT CARD**

Email Receipt: Y/N

Email Address: tom.johnson@email.com

SMS Notifications for Workout Reminders: Y/N

Additional Notes: Requires evening sessions only

MARK SCHEME:

- ***Date***
- ***Member ID***
- ***Member first name***
- ***Member last name***
- ***Trainer name***
- ***Membership type***
- ***Email***
- ***Email receipt***

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- ***Membership Start Date***
- ***Payment type***
- ***SMS notifications***
- ***Additional Notes***
- ***Workout Plan Details***

(1 mark per bullet point, up to a max of 8 marks. Any other reasonable implicit attributes should be accepted. Use your discretion.)

b) The gym is considering switching from a spreadsheet to a database system for managing memberships and training sessions. Identify **TWO (2)** advantages of using a database system compared to the current spreadsheet. **2**

MARK SCHEME:

Include but not limited to:
better management reporting, sharing of data, security, remote access.

1 mark for each, up to a max of 2 marks.

Any other suitable advantages should be accepted.

Total 10 Marks

Question 5

a) Flex Gym wants to create a database to manage its members, workout plans, and personal trainer assignments. **6**

Members can have multiple workout plans and trainers are assigned to members for personalised coaching.

Create a CRUD matrix to show the following transactions:

Transaction 1 – Add a new workout plan to the system.

Transaction 2 – Register a new gym member.

Transaction 3 – Assign a personal trainer to an existing gym member.

Transaction 4 – Update the details of a member's workout plan, i.e., change the duration.

Transaction 5 – Remove a workout plan that is no longer offered.

Transaction 6 – Retrieve a member's personal trainer assignment history.

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Transaction	Workout Plan	Member	Personal Trainer
1			
2			
3			
4			
5			
6			

MARK SCHEME:

Award 1 mark for each correct row in a table, up to a maximum of 6 marks. 0.5 mark should be given for a partially correct row. Marks should be rounded up at the end of the question.

Transaction	***Workout Plan***	***Member***	***Trainer Assignment***
T1	***C***		
T2		***C***	
T3			***C***
T4	***U***	***R***	
T5	***D***		
T6		***R***	***R***

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b) Identify **and** explain **TWO (2)** potential issues that could arise if data integrity is not enforced when removing a workout plan that is currently in use. **4**

MARK SCHEME:

- ***Disconnected Records: Deleting a workout plan without removing linked member records could lead to inaccessible data.***
- ***Inconsistent States: The system might reference workout plans that no longer exist.***

1 mark for each suitable issue above, and 1 mark for its explanation, up to a max of 4 marks.
Any other suitable potential issues should be accepted.

Total 10 Marks

Question 6

Consider the following scenario.

A hotel management system uses a database to manage rooms, guests and reservations.

Entities:

- Room
- Guest
- Reservation

Write the appropriate SQL command to perform each of the following tasks to the room data provided below.

room

room_id	type	price_per_night (£)	is_occupied
101	Deluxe	120.00	FALSE
102	Standard	75.00	TRUE
103	Suite	137.50	FALSE

a) Create the `room` table. **5**

MARK SCHEME:

```
CREATE TABLE room ( (1 mark)
    room_id INT PRIMARY KEY, (1 mark)
    type VARCHAR(50), (1 mark)
    price_per_night DECIMAL (7,2), (1 mark)
    is_occupied BOOLEAN (1 mark)
);
```


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b) Update the `room` table to include a new column named `is_cleaned` to track whether a room is cleaned or not. **2**

MARK SCHEME:

ALTER TABLE room (1 mark)
ADD COLUMN is_cleaned BOOLEAN; (1 mark)

c) The `price_per_night` for the room with `room_id` 102 needs to be updated to £80.00. **3**

MARK SCHEME:

UPDATE room (1 mark)
SET price_per_night = 80.00 (1 mark)
WHERE room_id = 102; (1 mark)

Total 10 Marks

Question 7
The hotel management database system contains the following tables.

room

room_id	type	price_per_night (£)	is_occupied
101	Deluxe	120.00	FALSE
102	Standard	75.00	TRUE
103	Suite	137.00	FALSE

guest

guest_id	first_name	last_name	contact_number
201	Chris	Lee	123456
202	Alice	Smith	234567
302	Aidan	Johnson	345678

reservation

reservation_id	check_in	check_out	guest_id(FK)	room_id(FK)
301	2025-12-23 09:00:00	2025-12-23 10:00:00	201	101
302	2025-04-02 12:00:00	2025-04-10 14:10:00	202	102
303	2025-10-14 08:00:00	2025-10-14 11:30:00	203	103
304	2025-09-20 19:00:00	2025-09-20 22:00:00	201	103
305	2025-09-16 11:00:00	2025-09-16 13:00:00	202	101

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a) Write an SQL command to list all the reservations for rooms with the type “suite”. **3**

MARK SCHEME:

SELECT * FROM reservation (1 mark)

JOIN room ON reservation.room_id = room.room_id (1 mark)

WHERE room.type = 'Suite'; (1 mark)

b) Write an SQL command to display each guest’s name, contact number and the total number of reservations for each guest who has made more than ONE (1) reservation. **4**

MARK SCHEME:

SELECT guest.first_name, guest.last_name, guest.contact_number,
COUNT(*) AS total_reservations
FROM reservation (1 mark)
JOIN guest ON reservation.guest_id = guest.guest_id (1 mark)
GROUP BY guest.guest_id (1 mark)
HAVING COUNT(*) > 1; (1 mark)

c) Write an SQL command to list the guest names and the room type for all guests who have reserved a room. **3**

MARK SCHEME:

SELECT guest.first_name, guest.last_name, room.type
FROM reservation (1 mark)
JOIN guest ON reservation.guest_id = guest.guest_id (1 mark)
JOIN room ON reservation.room_id = room.room_id; (1 mark)

Total 10 Marks

Question 8

Identify **and** explain **FIVE (5)** reasons why an SME (Small or Medium Enterprise) might decide to implement a database system to replace their current manual paper-based record keeping system. **10**

MARK SCHEME:

Suggested answers are not limited to:

- ***Improve data accuracy***
- ***Efficient data processing/retrieval***
- ***Automatic calculation***
- ***Centralised data access for staff***
- ***Better data security***

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- ***Handle large amount of data efficiently***
- ***Backup***
- ***Automated report generation***
- ***Natural disasters recovery***

1 mark for each correctly identified advantage and 1 mark for each example, up to a maximum of 10 marks.

Any suitable advantages and examples should be accepted.

Total 10 Marks

Question 9

a) A national healthcare provider relies on a central database to manage critical patient information, including medical records, treatment history and billing details. However, during a recent server failure, it was discovered that the database had no recent backups and the recovery processes were inadequate. **8**

Discuss **FOUR (4)** key areas where the lack of proper database backup and recovery processes can negatively impact the healthcare provider. For each area, include a specific example of how the failure could affect patient care or business operations.

MARK SCHEME:

Data Loss (1 mark):

Example: A patient's surgical history may be irretrievable, leading to incorrect or delayed treatment decisions. (1 mark)

Availability Loss (1 mark):

Example: Doctors may not access patient information for urgent care, delaying surgeries or treatments. (1 mark)

Legal and Regulatory Penalties (1 mark):

Example: A data audit reveals no backup strategy, leading to legal action and financial penalties for the organisation.

Reputation Damage (1 mark):

Example: News of data loss spreads publicly, causing patients to switch to providers with better data management practices. (1 mark)

1 mark for each identified key area and 1 mark for example, up to a max of 8 marks.

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b) A local restaurant chain is planning to expand its operations and implement a centralised database to manage inventory, orders and customer information across all its branches. The management has limited technical expertise and prefers a solution that minimises the need for managing database infrastructure. **2**

Identify a suitable cloud database model for them. Provide **ONE (1)** reason to support your answer.

MARK SCHEME:

Managed (1 mark). Reduce the need for in-house database expertise (1 mark), allowing the restaurant to focus on services improvement rather than database maintenance (1 mark). Ideal for a restaurant with limited technical resources to handle database tasks (1 mark). Managed models offer scalability, which allows the restaurant to grow (1 mark).

1 mark for the correct cloud database model.
1 mark for any one of the reasons above.
Any other suitable reasons should be accepted

Total 10 Marks

Question 10

A global online multiplayer gaming platform is building a backend system to manage its diverse data requirements.

The platform needs to store player profiles, in-game leader boards, real-time matchmaking data and social connections between players.

To meet these needs, the platform is considering adopting **NoSQL** databases for efficient scalability and performance.

a) Define what NoSQL stands for. **1**

MARK SCHEME:

Non SQL/Not Only SQL/Non-relational SQL.

1 mark for any one above.

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b) Identify **THREE (3)** NoSQL data models that could be used to store different types of data for the platform. Provide **ONE (1)** example of data that each model would handle best. **6**

MARK SCHEME:

- ***Key-Value Store (1 mark): Session data or temporary matchmaking preferences for players, such as region, skill level or preferred game mode. (1 mark)***
- ***Document Store (1 mark): Player profiles, including attributes like username, inventory items, in-game achievements and settings. (1 mark)***
- ***Graph Database (1 mark): Social connections between players, such as friend lists, memberships or team compositions. (1 mark)***

1 mark for each suitable data model, up to a maximum of 3 marks.

1 mark for each correct data, up to a maximum of 3 marks.

Any other suitable answers should be accepted.

c) Explain how the 3V model (Volume, Variety and Velocity) applies to the data management challenges the online multiplayer gaming platform might face when handling large-scale data. You must explain each V in this context. **3**

MARK SCHEME:

- ***Volume:***

  Refers to the large amount of data generated by players during gameplay. (1 mark)

- ***Variety:***

  Describes the different types of data produced by the gaming platform. (1 mark)

- ***Velocity:***

  Refers to the speed at which data is generated and processed during gameplay. (1 mark)

1 mark for each brief explanation, up to a max of 3 marks.

Any other suitable answers should be accepted.

Total 10 Marks

End of Paper

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Learning Outcomes matrix

Question	Learning Outcomes assessed	Marker can differentiate between varying levels of achievement
1	1,5	Yes
2	4	Yes
3	3	Yes
4	1, 4	Yes
5	4	Yes
6	5	Yes
7	5	Yes
8	1, 2	Yes
9	1, 7	Yes
10	6	Yes

Grade descriptors

Learning Outcome	Fail	Referral	Pass	Merit	Distinction
Understand the concepts associated with database systems	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.

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Understand the concepts associated with the relational model	Has little to no awareness of principles and concepts underlying theoretical frameworks and approaches and is unable to identify associated strengths or weaknesses.	Has vague awareness of principles and concepts underlying theoretical frameworks and approaches and is only superficially able to identify associated strengths or weaknesses.	Has satisfactory awareness of principles and concepts underlying theoretical frameworks and approaches and demonstrates sufficient ability to identify associated strengths and weaknesses.	Has very good awareness of principles and concepts underlying theoretical frameworks and approaches and is able in detail to identify associated strengths and weaknesses.	Has an excellent awareness of principles and concepts underlying theoretical frameworks and approaches and is comprehensively able to identify associated strengths and weaknesses.
Understand how to design and develop a database system	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Be able to develop and enhance a logical database design	Demonstrates little to no ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates a very good ability to use design principles to effectively develop an artefact to solve an identified issue.	Demonstrates an excellent ability to use design principles to effectively develop an artefact to solve an identified issue.
Be able to develop and enhance a database system using SQL	Demonstrates little to no ability to adequately develop an artefact to solve an identified issue.	Demonstrates an extremely limited ability to adequately develop an artefact to solve an identified issue.	Demonstrates a satisfactory ability to develop an artefact to solve an identified issue.	Demonstrates a very good ability to develop an artefact to solve an identified issue.	Demonstrates an excellent ability to develop an artefact to solve an identified issue.

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Understand the principles behind Big Data and the NoSQL databases	Displays no understanding of Big Data and NoSQL databases. Unable to recognise the key features and benefits of NoSQL over traditional databases or how Big Data impacts database design and scalability.	Shows a basic awareness of Big Data and NoSQL databases. Recognises some key aspects but fails to understand the broader implications and technical considerations involved.	Demonstrates satisfactory comprehension of Big Data principles and the operational basics of NoSQL databases. Can distinguish NoSQL databases from traditional databases and understands their use cases.	Exhibits a strong grasp of the principles behind Big Data and NoSQL databases. Can articulate the advantages of NoSQL databases in handling large datasets and different data types.	Has an in-depth understanding of Big Data and NoSQL databases. Can critically evaluate and recommend NoSQL database solutions for complex Big Data problems, demonstrating a sophisticated understanding of scalability and performance optimisation.
Understand the concepts and principles of Cloud database	Lacks understanding of the fundamental concepts and principles of Cloud databases. Cannot identify the main advantages or the operational model of Cloud databases.	Has a superficial grasp of Cloud databases. Recognises some concepts but cannot articulate how Cloud databases differ from on-premises solutions or their benefits.	Has a solid understanding of Cloud database concepts and principles. Understands the benefits of scalability, availability, and managed services provided by Cloud databases.	Possesses a very good understanding of Cloud databases. Can discuss the importance of Cloud databases in modern architectures and understands their significance in providing flexible and scalable data storage solutions.	Demonstrates an excellent and comprehensive understanding of the key concepts and principles of Cloud databases. Can describe in detail how Cloud databases offer advantages in terms of accessibility, maintenance, and scalability, and can suggest appropriate use cases for Cloud databases.

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